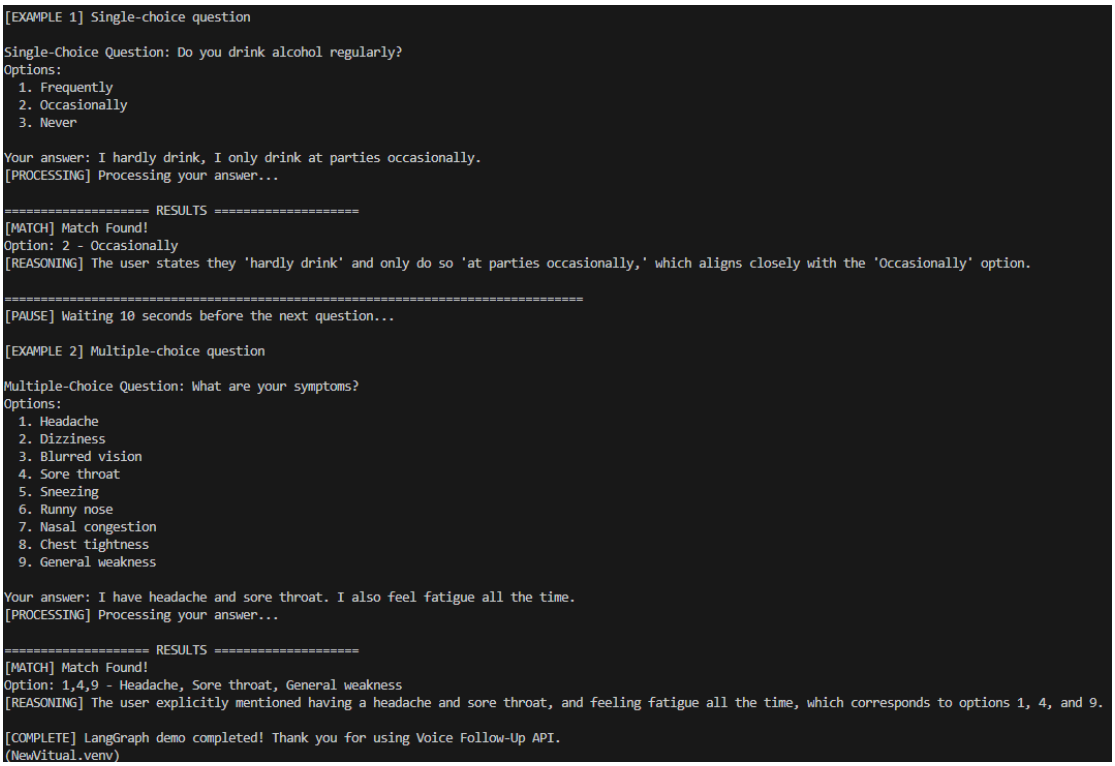
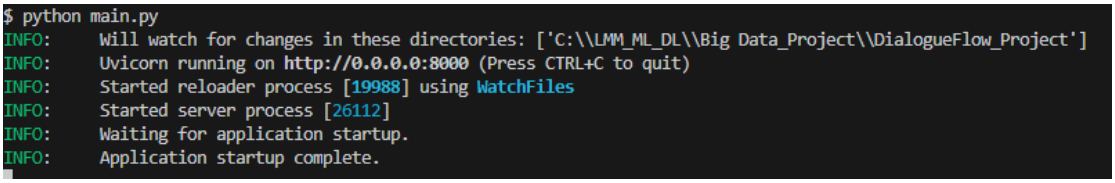
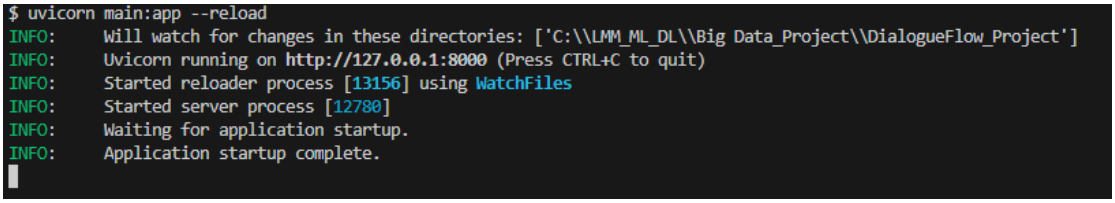
Following screenshots are program results for codes that we conduct:

For LangGraph Dialogue Followup (Dialogue_Followup_Langgraph.py):



For FastAPI of LangGraph Program (notice this is achieved by main.py):





Screenshots for Interact API file (https://127.0.0.1:8000/docs):

Dialogue Follow-Up API 1.0.0 OAS 3.1

/openapi.json

default

GET / Root

Parameters | Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: text/html

Controls Accept header.

Example Value | Schema

```
string
```

GET /health Health Check

Parameters | Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{
  "status": "string",
  "timestamp": "string",
  "version": "string"
}
```

POST /process-dialogue Process Dialogue

Parameters | Try it out

No parameters

Request body required — application/json

Example Value | Schema

```
{
  "question": "string",
  "options": [
    {
      "opt_code": "string",
      "opt_content": "string"
    }
  ],
  "user_answer": "string",
  "question_type": "single",
  "api_key": "string",
  "model": "Qwen/Qwen2.5-Coder-32B-Instruct"
}
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{
  "success": true,
  "timestamp": "string",
  "question": "string",
  "user_answer": "string",
  "question_type": "string",
  "match_result": "string",
  "error_message": "string",
  "processing_time_ms": 0
}
```

422	Validation Error	*No links*

Media type: application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

POST /demo/single Demo Single

Parameters | Try it out

No parameters

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type: application/json

Controls Accept header.

Example Value | Schema

```
{
  "success": true,
  "timestamp": "string",
  "question": "string",
  "user_answer": "string",
  "question_type": "string",
  "match_result": "string",
  "error_message": "string",
  "processing_time_ms": 0
}
```

POST /demo/multiple Demo Multiple

Parameters | Try it out

No parameters

For calling the new generated API to test its function (Dialogue_Followup_FastAPI_Test.py):

```
$ python main.py
INFO:     Will watch for changes in these directories: ['C:\\LMM_ML_DL\\Big Data_Project\\DialogueFlow_Project']
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [16832] using WatchFiles
INFO:     Started server process [26932]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
```

Following is result of single choice question test:

```
$ python Dialogue_Followup_FastAPI_Test.py
Local FastAPI Dialogue Client
==============================
[SUCCESS] API server is healthy!

Choose question type:
1. Single Choice Question
2. Multiple Choice Question
Enter your choice (1 or 2): 1

Single Choice Question: How often do you exercise?
Options (select one):
  1. Daily
  2. Weekly
  3. Monthly
  4. Rarely

Enter your answer in natural language:
(Example: 'I exercise every day' or 'Almost never')
Your answer: I do exercise every day.

[PROCESSING] You answered: 'I do exercise every day.'
[INFO] Processing as single choice question...
[INFO] Sending to API for processing...

================================================
DIALOGUE PROCESSING RESULT
================================================
Success: True
Question: How often do you exercise?
User Answer: I do exercise every day.

[MATCH FOUND]
Option: 1 - Daily
Reasoning: The user's answer 'I do exercise every day' directly corresponds to the option 'Daily', indicating a high level of confidence in this match.
(NewVitual.venv)
```

Following is result of multiple choice question with several answers test:

```
$ python Dialogue_Followup_FastAPI_Test.py
Local FastAPI Dialogue Client
==============================
[SUCCESS] API server is healthy!

Choose question type:
1. Single Choice Question
2. Multiple Choice Question
Enter your choice (1 or 2): 2

Multiple Choice Question: What activities do you enjoy doing in your free time?
Options (you can select multiple):
  A. Reading books
  B. Playing sports
  C. Watching movies
  D. Cooking
  E. Traveling

Enter your answer in natural language:
(Example: 'I like reading and watching movies' or 'Sports and cooking are my favorites')
Your answer: I prefer travelling and I cook from time to time.

[PROCESSING] You answered: 'I prefer travelling and I cook from time to time.'
[INFO] Processing as multiple choice question...
[INFO] Sending to API for processing...

================================================
DIALOGUE PROCESSING RESULT
================================================
Success: True
Question: What activities do you enjoy doing in your free time?
User Answer: I prefer travelling and I cook from time to time.

[MATCHES FOUND] (2 options selected)
1. Option: D - Cooking
   Reasoning: The user mentioned cooking from time to time, which aligns with this option.
2. Option: E - Traveling
   Reasoning: The user explicitly stated they prefer traveling, which directly matches this option.
(NewVitual.venv)
```